

Supply Chain Performance & Analytics Dashboard

Data Profiling & Validation Report

Version: Final

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Date: 05-Sep-2026

1. Data Model Overview

The project uses CSV datasets covering key areas of the supply chain, including procurement, inventory, production, and sales.

The datasets were hosted in Supabase / PostgreSQL and queried using SQL for data exploration, validation, and business analysis.

The data is organized into dimension tables and fact tables to support the analytical data model.

2. Dataset Structure

Dimension Tables

Dataset	Rows	Columns	Purpose
dim_product.csv	16	11	Product details and classifications
dim_customer.csv	5	7	Customer information
dim_facility.csv	6	8	Facility information
dim_supplier.csv	7	8	Supplier information
dim_date(3).csv	731	11	Date and calendar attributes

Fact Tables

Dataset	Rows	Columns	Purpose
fact_inventory.csv	1,152	7	Inventory levels and reorder points
fact_production.csv	4,500	8	Production output and defective units
fact_sales.csv	8,500	14	Sales, revenue, quantity, and profit
fact_procurement.csv	2,200	11	Procurement quantities, costs, lead times, and quality

3. Dimension Table Validation

The dimension tables were checked for:

- Missing values
- Duplicate records
- Unique IDs
- Key integrity

The validation showed that the dimension tables contained no missing values or duplicate records, and their identifier fields were unique.

The dimension tables were therefore suitable for use in the integrated analytical model.

4. Fact Table Validation

The procurement, inventory, production, and sales fact tables were checked for:

- Missing values
- Duplicate records
- Foreign key integrity
- Numeric consistency
- Business-rule consistency

No missing values or duplicate records were identified in these fact tables, and the relevant foreign keys were consistent with the available dimension tables.

5. Data Quality Checks

5.1 Missing Values

The datasets were checked for null or missing values.

Result:

No missing values were identified in the validated dimension and fact tables.

5.2 Duplicate Records

The tables were checked for duplicate records.

Result:

No duplicate records were identified in the validated dimension and fact tables.

5.3 Primary Key / ID Validation

Dimension table identifiers were checked for uniqueness.

Result:

The dimension IDs were unique and suitable for establishing relationships with the fact tables.

5.4 Foreign Key Validation

Foreign key relationships in the procurement, inventory, production, and sales tables were checked against their corresponding dimension tables.

Result:

The validated fact tables showed consistent foreign key relationships.

6. Business Rule Validation

Several numerical relationships were checked to confirm data consistency.

Sales

Sales calculations were validated against the available revenue, quantity, and profit fields.

Result: The sales arithmetic checks were consistent.

Procurement

The relationship between order quantity, unit cost, and total cost was validated:

Total Cost = Order Quantity × Unit Cost

Result: The procurement total cost calculation matched the expected relationship.

Production

Production quantity and defective units were reviewed to validate the aggregate defect rate.

Result: Minor row-level rounding differences were observed, but the aggregate defect rate was considered suitable for analysis.

7. Inventory Validation

The inventory data was reviewed to identify the latest available inventory snapshot.

The latest snapshot contained 16 product records per facility for the relevant facilities.

Inventory levels were compared with reorder points to identify products and facilities with potential inventory risk.

This validation supported the use of:

- Current Inventory
- Inventory Risk Rate
- Inventory Shortfall
- Estimated Inventory Value

for the final analysis.

8. Shipment Data Validation

The shipment dataset was reviewed but was not included in the final integrated model.

The validation identified inconsistencies between its IDs and the available dimension tables:

- Shipment product IDs extended beyond the available product IDs.
- Shipment customer IDs extended beyond the available customer IDs.
- Shipment facility IDs extended beyond the available facility IDs.

Because these identifiers could not be reliably matched to the existing dimensions, the shipment table was excluded rather than arbitrarily replacing or remapping its IDs.

9. Final Data Model Scope

After validation, the final integrated model includes:

Dimensions

- dim_product
- dim_customer
- dim_facility
- dim_supplier
- dim_date

Facts

- fact_procurement
- fact_inventory
- fact_production
- fact_sales

The fact_shipment table was excluded from the integrated model due to ID inconsistencies.

10. Data Preparation & Validation Workflow

The data validation process followed this workflow:

CSV Files → Supabase / PostgreSQL → Data Profiling → Data Quality Checks → Relationship Validation → Business Rule Validation → Final Data Model → SQL Analysis → Power BI

11. Conclusion

The validation confirmed that the data was suitable for supply chain analysis and Power BI reporting. Shipment and capacity-related limitations were excluded to avoid unreliable results. The final model focuses on procurement, inventory, production, and sales performance.